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### Techniques for processing devices

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#### Abstract

Representative techniques provide process steps for forming a microelectronic assembly, including preparing microelectronic components such as dies, wafers, substrates, and the like, for bonding. One or more surfaces of the microelectronic components are formed and prepared as bonding surfaces. The microelectronic components are stacked and bonded without adhesive at the prepared bonding surfaces.

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**Inventors:** Uzoh; Cyprian Emeka (San Jose, CA), Mirkarimi; Laura Wills (Sunol, CA), Gao; Guilian (Campbell, CA), Fountain, Jr.; Gaius Gillman (Youngsville, NC)

**Applicant:** ADEIA SEMICONDUCTOR BONDING TECHNOLOGIES INC. (San Jose, CA)

**Family ID:** 67541023

**Assignee:** ADEIA SEMICONDUCTOR BONDING TECHNOLOGIES INC. (San Jose, CA)

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## Field of Classification Search

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*Primary Examiner:* Kebede; Brook

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## **Background/Summary**

**PRIORITY CLAIM AND CROSS-REFERENCE TO ANY RELATED APPLICATIONS** (1) This application is a continuation of U.S. Non-Provisional application Ser. No. 17/344,100, filed Jun. 10, 2021, which is a continuation of U.S. Non-Provisional application Ser. No. 16/919,989, filed Jul. 2, 2020, which is a continuation of U.S. Non-Provisional application Ser. No. 16/262,489, filed Jan. 30, 2019, which claims priority to U.S. Provisional Application No. 62/631,216, filed Feb. 15, 2018, which are hereby incorporated by reference in their entirety.

### **FIELD**

(1) The following description relates to integrated circuits (“ICs”). More particularly, the following description relates to manufacturing IC dies and wafers.

### **BACKGROUND**

(2) Microelectronic elements often comprise a thin slab of a semiconductor material, such as silicon or gallium arsenide, commonly called a semiconductor wafer. A wafer can be formed to include multiple integrated chips or dies on a surface of the wafer and/or partly embedded within the wafer. Dies that are separated from a wafer are commonly provided as individual, prepackaged units. In some package designs, the die is mounted to a substrate or a chip carrier, which is in turn mounted on a circuit panel, such as a printed circuit board (PCB). For example, many dies are provided in packages suitable for surface mounting.

(3) Packaged semiconductor dies can also be provided in “stacked” arrangements, wherein one package is provided, for example, on a circuit board or other carrier, and another package is mounted on top of the first package. These arrangements can allow a number of different dies to be mounted within a single footprint on a circuit board and can further facilitate high-speed operation by providing a short interconnection between the packages. Often, this interconnect distance can be only slightly larger than the thickness of the die itself. For interconnection to be achieved within a stack of die packages, interconnection structures for mechanical and electrical connection may be provided on both sides (e.g., faces) of each die package (except for the topmost package).

(4) Additionally, dies or wafers may be stacked in a three-dimensional arrangement as part of various microelectronic packaging schemes. This can include stacking a layer of one or more dies, devices, and/or wafers on a larger base die, device, wafer, substrate, or the like, stacking multiple dies or wafers in a vertical or horizontal arrangement, and various combinations of both. Dies or wafers may be bonded in a stacked arrangement using various bonding techniques, including direct dielectric bonding, non-adhesive techniques, such as ZiBond® or a hybrid bonding technique, such as DBI®, both available from Invensas Bonding Technologies, Inc. (formerly Ziptronix, Inc.), an Xperi company (see for example, U.S. Pat. Nos. 6,864,585 and 7,485,968, which are incorporated herein in their entirety).

(5) There can be a variety of challenges to implementing stacked die and wafer arrangements.

When bonding stacked dies using a direct bonding or hybrid bonding technique, it is usually desirable that the surfaces of the dies to be bonded be extremely flat, smooth, and clean. For instance, in general, the surfaces should have a very low variance in surface topology and have a low level of impurities, particles, or other residue. The removal of particles or residues can improve the cleanliness and flatness of the surfaces and the reliability of the bond between the layers, however, the removal of particles and residue can sometimes be problematic.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) The detailed description is set forth with reference to the accompanying figures. In the figures, the left-most digit(s) of a reference number identifies the figure in which the reference number first appears. The use of the same reference numbers in different figures indicates similar or identical items.

(2) For this discussion, the devices and systems illustrated in the figures are shown as having a multiplicity of components. Various implementations of devices and/or systems, as described herein, may include fewer components and remain within the scope of the disclosure. Alternatively, other implementations of devices and/or systems may include additional components, or various combinations of the described components, and remain within the scope of the disclosure.

(3) FIG. 1A shows an example profile view of a substrate and two dies prior to bonding the dies to the substrate.

(4) FIG. 1B shows an example profile view of the substrate and the two dies after bonding the dies to the substrate.

(5) FIG. 1C shows an example profile view of the substrate and bonding multiple dies to the substrate.

(6) FIG. 2 is a flow diagram illustrating an example process for preparing a substrate for bonding, according to an embodiment.

(7) FIG. 3 is a flow diagram illustrating an example process for preparing dies for bonding, according to an embodiment.

(8) FIG. 4 is a flow diagram illustrating another example process for preparing dies for bonding, according to an embodiment.

(9) FIG. 5 is a flow diagram illustrating another example process for preparing dies for bonding, according to an embodiment.

(10) FIG. 6A is a flow diagram illustrating an example process for preparing a substrate for bonding, according to an embodiment.

(11) FIG. 6B is a flow diagram illustrating an example process for preparing dies for bonding, according to an embodiment.

(12) FIG. 7A is a flow diagram illustrating another example process for preparing a substrate for bonding, according to an embodiment.

(13) FIG. 7B is a flow diagram illustrating another example process for preparing dies for bonding, according to an embodiment.

### SUMMARY

(14) Representative techniques and devices are disclosed, including process steps for forming a novel microelectronic assembly. Processes include preparing microelectronic components such as dies, wafers, substrates, and the like, for bonding. In various implementations, one or more surfaces of the microelectronic components are formed and prepared as bonding surfaces. The microelectronic components may be stacked and bonded without adhesive at the prepared bonding surfaces.

(15) In various implementations, a method for forming a microelectronic assembly includes

preparing a bonding surface of a first substrate, including: planarizing the bonding surface of the first substrate to have a first predetermined minimal surface variance and plasma activating the bonding surface of the first substrate. The method further includes preparing a first bonding surface of a second substrate, including planarizing the first bonding surface of the second substrate to have a second predetermined minimal surface variance.

(16) In an implementation, the method includes mounting the second substrate to a dicing sheet or dicing tape held by a dicing frame and singulating the second substrate into a plurality of dies while the second substrate is mounted to the dicing sheet or dicing tape. Each die of the plurality of dies has a first bonding surface comprised of a portion of the first bonding surface of the second substrate.

(17) The method may include processing the plurality of dies while the plurality of dies is mounted to the dicing sheet or dicing tape, selecting a die of the plurality of dies using a pick and place tool, and directly bonding the first bonding surface of the die to the bonding surface of the first substrate without adhesive, and without activating the first bonding surface of the die.

(18) In additional implementations, techniques and methods include preparing a back side of the second substrate or the plurality of dies (or the single die) for bonding an additional die (or multiple additional dies) to the bonded die.

(19) In some implementations, unique sets of processing steps are used to clean the bonding surfaces of the first and/or second substrates, or the die. For example, in some embodiments, one or more unique formulary may be used for cleaning. In other embodiments, a megasonic transducer, and/or a mechanical brush may be used to assist in cleaning.

(20) In an alternate implementation, the bonding surfaces of one or more of the substrates or dies may be coated with one or more protective coatings during processing. For example, in one embodiment, individual protective coatings may render the bonding surfaces hydrophobic or hydrophilic. The coatings may prevent contamination to bonding surfaces, protect against damage to the bonding surfaces during processing, or the like. One or more of the coatings may be removed during other process steps. Additionally, one or more of the processing tools that handle components may be coated or formed to be hydrophobic as a technique for preventing contamination of clean components during handling.

(21) Various implementations and arrangements are discussed with reference to electrical and electronics components and varied carriers. While specific components (i.e., dies, wafers, integrated circuit (IC) chip dies, substrates, etc.) are mentioned, this is not intended to be limiting, and is for case of discussion and illustrative convenience. The techniques and devices discussed with reference to a wafer, die, substrate, or the like, are applicable to any type or number of electrical components, circuits (e.g., integrated circuits (IC), mixed circuits, ASICs, memory devices, processors, etc.), groups of components, packaged components, structures (e.g., wafers, panels, boards, PCBs, etc.), and the like, that may be coupled to interface with each other, with external circuits, systems, carriers, and the like. Each of these different components, circuits, groups, packages, structures, and the like, can be generically referred to as a “microelectronic component.” For simplicity, unless otherwise specified, components being bonded to another component will be referred to herein as a “die,” and the other component that the die is bonded to will be referred to herein as a “substrate.”

(22) Implementations are explained in more detail below using a plurality of examples. Although various implementations and examples are discussed here and below, further implementations and examples may be possible by combining the features and elements of individual implementations and examples.

## DETAILED DESCRIPTION

(23) Overview

(24) Referring to FIGS. 1A-1C, in various examples, techniques for processing devices to be stacked and directly or intimately bonded are disclosed. The disclosure describes example

processes that include bonding one or more dies **102** to a substrate **104**. However, the processes can also be used to bond wafers to wafers, dies to dies, dies to wafers, and so forth. The diagrams in the accompanying figures show some processes for forming and preparing a substrate **104** (and particularly a bonding surface **108** of a substrate **104**) for bonding, and other processes for forming and preparing the one or more dies **102** (and particularly the bonding surfaces **106** and **110** of the dies **102**) for bonding to the substrate **104** and to other dies **102**. At the end of these processes, the one or more dies **102** are bonded (directly bonded, for example, without an adhesive) to the substrate **104** or to another die **102**. The bonding is a spontaneous process that takes place at ambient conditions when the two surfaces (**106** and **108** and/or **106** and **110**) are brought together.

(25) Profile views showing example dies **102** and an example substrate **104** are illustrated at FIGS. **1A-1C**. FIG. **1A** illustrates dies **102** and the substrate **104** prior to bonding, and FIG. **1B** shows the dies **102** and the substrate **104** after bonding. FIG. **1C** shows bonding multiple dies **102** to the substrate **104**, for instance, by stacking and bonding dies **102B** to dies **102A**, after bonding the dies **102A** to the substrate **104**. Additional dies **102** (C-N) may be stacked and bonded to the dies **102B** as desired in like manner.

(26) In various embodiments, an example die **102** or an example substrate **104** can include one or more conductive traces or interconnects (not shown) surrounded by an insulating material (oxide, for example) on the bonding surface (**106**, **110**) of the die **102** or the bonding surface **108** of the substrate **104**. For example, a die **102** or a substrate **104** can include multiple conductive and insulating layers (not shown) to determine the function (e.g., memory, processing, programmable logic, etc.) of the die **102** or the substrate **104**. Exposed interconnects on the bonding surface (**106**, **108**, or **110**) can be electrically continuous with the various conductive layers, and provide an interface for the die **102** or the substrate **104**.

(27) When bonding the dies **102** to the substrate **104** (or bonding dies to dies, dies to wafers, wafers to wafers, etc.) using a direct bonding or hybrid bonding technique, it is usually desirable that the surfaces of the dies **102** and substrate **104** to be bonded be extremely flat and smooth. For instance, in general, the bonding surfaces (**106** and **108**, and **106** and **110**) should have a very low variance in surface topology (i.e., nanometer scale variance, for instance below 2 nm and preferably below 0.5 nm), so that the bonding surfaces (**106** and **108**, and **106** and **110**) can be closely mated to form a lasting bond. It is also usually desirable that the bonding surfaces (**106** and **108**, and **106** and **110**) be clean and have a low level of impurities, particles, or other residue that are large enough in size to cause bonding voids that can cause electrical continuity failures or other bonding defects.

(28) For instance, particles and residues remaining from processing steps can result in voids at the bonding interfaces between the stacked dies **102** and the substrate **104**. If the voids are substantially smaller than the metallic electrical interconnect size, they may be acceptable. However, particles that cause bonding defects in sizes that are close to or exceed the electrical interconnect size often cannot be tolerated. Also, temporarily bonding dies, wafers, and substrates for processing or handling (e.g., using polymeric or inorganic layers, etc. for temporary bonds) can also be problematic, since the temporary carriers and substrates can leave behind bonding residue when removed, which can persist even after cleaning.

(29) Residue from temporary adhesive layers, which can be comprised of high temperature polymers and the like, can be discontinuous with varying thicknesses on the die surfaces **106** and **110** or the substrate surface **108** (e.g., thickness may range from 10 nm to 50 um). Plasma cleaning, such as ashing with oxygen plasma, can be used to remove thin residue, but even long oxygen plasma ashing steps (e.g., over 40 minutes) may be ineffective to remove the thickest residues. Additionally, long oxygen ashing processes can tend to oxidize the conductive features on the bonding surfaces (**106**, **110**, and/or **108**) and reduce the planarity of the cleaned surfaces.

(30) For example, a bonding surface (**106**, **110**, and/or **108**) may include recessed conductive features, where the nominal conductive layer recess is about 1 nm to 20 nm below the surface of an adjacent dielectric layer or surface. After long oxygen ashing of the wafer or die surface, the

formerly recessed conductive features may change substantially to interfere with bonding or with formation of metal interconnects after bonding. For instance, the formerly recessed conductive features may now protrude over the surface of adjacent dielectric regions. The protrusion of the conductive features may vary from 3 nm to 30 nm in some cases, depending on the ashing process parameters. These conductive asperities over the bonding surface (**106**, **110**, and/or **108**) can hinder or prevent intimate mating of bonding layers or surfaces.

(31) In such cases, a high temperature (e.g., over 50° C.) wet process is sometimes used to remove thick residue; however, such a process may not be compatible with other die **102** or substrate **104** layers or materials. For instance, a high temperature wet process can degrade the polished metal layers of the die **102** or substrate **104**, reducing device yield. Further, the high temperatures may not be compatible with some die **102** or substrate **104** processing components (e.g., plastic dicing tapes, grip rings, etc.).

(32) In some conventional wafer cleaning processes, wet chemical cleaning solutions, including aggressive cleaning solutions may be effective in removing organic residues from surfaces of wafers. For example, cleaning wafers with mixtures of hydrogen peroxide in a solution of ammonium hydroxides, sulfuric acid, or hydrochloric acids, and their various combinations, at temperatures between 50 and 80° C. can produce pristine wafer surfaces. However, such aggressive cleaning steps would generally be impractical for cleaning the bonding surfaces **106** and **110** of the dies **102**, since these aggressive wet chemistries will not only clean the polymeric residues and particles, they may also dissolve large portions of the practical conductive features, which are often comprised of materials such as silver, copper, nickel, and their various alloys.

(33) Representative Processes

(34) FIGS. 2-7B illustrate representative processes **200-750** for forming and preparing microelectronic components (such as dies **102** and substrates **104**, for example) for bonding, such as for direct bonding without adhesive. The processes **200-750** include providing a bonding surface (such as bonding surfaces **106**, **110**, and **108**, for example) on the microelectronic components, or two bonding surfaces in some examples (such as bonding surfaces **106** and **110** on a single die **102**), planarizing the bonding surfaces, cleaning and activating (in some examples) the bonding surfaces, and the like.

(35) The order in which the processes **200-750** are described is not intended to be construed as limiting, and any number of the described process blocks in any of the processes **200-750** can be combined in any order to implement the processes, or alternate processes. Additionally, individual blocks may be deleted from any of the processes without departing from the spirit and scope of the subject matter described herein. Furthermore, the processes **200-750** can be implemented in any suitable hardware, software, firmware, or a combination thereof, without departing from the scope of the subject matter described herein.

(36) In alternate implementations, other techniques may be included in the processes **200-750** in various combinations and remain within the scope of the disclosure.

(37) Example Technique for Forming and Preparing the Substrate

(38) An example substrate **104** forming process **200** is illustrated at FIG. 2. In various embodiments, the substrate **104** may be comprised of silicon, germanium, a dielectric surface, direct or indirect gap semiconductor materials or layers or another suitable material. At block **202**, the process includes fabricating the substrate **104**. Briefly, forming the substrate **104** may include fabrication of devices in the substrate **104** such as the front end of the line (FEOL), multilayer backend of the line (BEOL) and other structures of interest, cleaning of the surface of the substrate **104**, and so forth.

(39) At block **204**, the process includes forming a direct bonding layer on a first (front) surface **108** of the substrate **104**. The materials for the bonding layer may be deposited or formed on the first surface **108**, and may be comprised of an inorganic dielectric material layer such as oxide, nitride, oxynitride, oxycarbide, carbides, nitrocarbides, diamond, diamond like materials, glasses, ceramics,

glass-ceramics, and the like, or a combination of an inorganic dielectric layer and one or more metal features. In some embodiments, the metal features can be recessed from the dielectric surface slightly, e.g., 1 nm to 20 nm below the surface of the dielectric. In some embodiments, forming the direct bonding layer can be incorporated into the wafer fabrication process (such as block **202**, for example) as the last metal layer formed. In one embodiment, the planar surface of the last metallized dielectric layer may comprise the bonding surface **108** and additional metallized dielectric coating to form the bonding layer **108** may not be necessary.

(40) Forming the direct bonding layer includes finishing the first surface **108** (i.e., bonding surface) to meet dielectric roughness specifications and metallic layer (e.g., copper, etc.) recess specifications to prepare the surface **108** for direct bonding. In other words, the bonding surface **108** is formed to be as flat and smooth as possible, with very minimal surface topology variance. Various conventional processes, such as chemical mechanical polishing (CMP) may be used to achieve the low surface roughness. The metallic layer may be configured to provide an electrical and/or thermal path or may instead be configured to balance out the metallization, through the use of so-called dummy pads, traces, patterns or the like.

(41) At block **206**, the bonding surface **108** of the substrate **104** is cleaned and/or activated in preparation for direct bonding, as described below.

(42) Example Technique for Forming and Preparing the Dies

(43) FIG. **3** illustrates an example process **300** for forming single-sided or double sided dies **102** from a wafer (which also may be referred to as a “second substrate,” and which may be comprised of similar or same materials and formed with similar or same processes and techniques as described above with reference to the substrate **104**). In an implementation, the process steps **302** and **304** of fabricating the wafer and forming the (front) bonding surface **106**, with the surface topology requirements of the bonding surface, are essentially the same as described above with reference to the substrate **104**, at blocks **202** and **204** of process **200**.

(44) At block **306**, after planarizing the bonding surface **106** of the wafer (e.g., by a CMP process, for example) to achieve a desired topology, a protective coating (such as a resist or other suitable material) is applied to the bonding surface of the wafer to protect the bonding surface **106** against contamination, to protect the exposed metallic layer against corrosion, and to protect the bonding surface **106** during singulation operations, which can tend to generate debris on the front and side surfaces of the dies **102**.

(45) Additionally or alternatively, the back surface of the wafer may be processed (e.g. through-die conductor reveal, planarization, etc.), forming a back bonding surface **110** on the dies **102**, which may also be coated with a protective coating, or the like.

(46) In some embodiments, more than one type of protective layer may be applied to the wafer surface. For example, the first protective layer may comprise a hydrophobic protective layer and overlying the hydrophobic layer may be a hydrophilic protective layer. In the example, the underlying hydrophobic layer allows for the use of aggressive etching chemicals during cleaning of the side surfaces of the dies **102** and also increases the shelf life of the prepared surfaces **106** and **110**. In some cases, the overlying hydrophilic layer receives or is impregnated with particles, debris, dicing tape, adhesive, etc. generated during the dicing process. The hydrophilic layer, along with the particles and debris, is removed. The hydrophobic layer may be temporarily maintained intact to protect the surface **106** (and/or **110**) during subsequent processing or storage.

(47) In some instances, as shown at block **308**, the wafer is thinned from the non-protected side to achieve a desired thickness. At block **310** the wafer is mounted to a dicing tape on a frame and singulated (block **312**) to form the dies **102**. The wafer may be singulated into a quantity of dies **102** using a saw, a laser (e.g. stealth laser), a plasma etching process, or other suitable technique. In various examples, the dies **102** are singulated while the wafer is mounted to a dicing sheet or dicing layer or the dicing tape (or the like), held by the frame.

(48) After dicing, the diced wafer on the dicing tape can be expanded and mounted into a grip ring



if desired. In various examples, the frame or grip ring may be comprised of plastic parts, metal parts, combinations of the same, and the like. The dicing tape can be any type of dicing tape commonly used in the industry. One example dicing tape includes a UV release tape. In some examples, the dies **102** may be shipped while on the grip ring or dicing frame.

(49) Forming and Preparing the Dies—Double-Sided Die Examples

(50) FIG. **4** illustrates an example process **400** for forming double-sided dies **102**, where both surfaces (**106** and **110**) of a die **102** will be bonded to substrates **104** or to other dies **102**, such as with multiple die-to-die or die-to-wafer applications (as shown in FIG. **1C**, for example). At block **402**, the process includes fabricating the wafer for the dies **102**, which may include fabricating through-silicon-vias (TSVs) at desired locations throughout the dies **102** for future through-device connections. In various embodiments, the wafer may be fabricated from silicon, germanium, or another suitable material.

(51) At block **404**, the process includes forming a direct bonding layer on a first (front) surface **106** of the dies **102**. The materials for the bonding layer may be deposited or formed on the first surface **106** and may be comprised of an inorganic dielectric material layer or a combination of an inorganic dielectric layer and one or more metal features, as described with reference to FIG. **2**. Similarly, as discussed earlier, the planar surface of the last metallized dielectric layer may comprise the bonding surface **106** and additional metallized dielectric coating to form the bonding layer **106** may not be necessary.

(52) Forming the direct bonding layer includes finishing the first surface **106** (i.e., bonding surface) to meet dielectric roughness specifications and metallic layer (e.g., copper, etc.) recess specifications to prepare the surface **106** for direct bonding. In other words, the bonding surface **106** is formed to be as flat and smooth as possible, with very minimal surface topology variance. Various conventional processes, such as chemical mechanical polishing (CMP) may be used to achieve the low surface roughness.

(53) In an implementation, at block **406** the prepared front surface of the wafer to be singulated into dies **102** is directly bonded to a support wafer (e.g., a “silicon carrier” in one example, but carriers of other materials may also be used), for handling the wafer during fabrication of the second (back) bonding surface **110**. In one example, the first surface of the wafer is bonded to a silicon wafer using a Zibond direct bonding technique, as described above.

(54) In various embodiments, directly bonding the silicon carrier to the wafer has multiple advantages. For instance, this technique removes any temperature limitations for back-side wafer processing. Further, the silicon carrier can have a similar or closely matched coefficient of thermal expansion (CTE), reducing or eliminating warping of the wafer during processing. This can improve the success and reliability of direct bonds with the wafer. Also, directly bonding the silicon carrier enables an ultra-thin wafer to be directly bonded to other dies, wafers, substrates, etc., where it would not be possible otherwise.

(55) At block **408**, prior to forming and polishing the second bonding surface **110**, the silicon substrate may be thinned, and the TSVs exposed. The second bonding surface **110** of the dies **102**, on the back side of the wafer, is formed and finished (as described above) to meet maximum dielectric roughness specifications and metallic layer (e.g., copper, etc.) recess specifications with minimal surface topology variance. Moreover, a protective coating may be applied to the polished second bonding surface **110**.

(56) At block **410**, the second surface **110** of the wafer may then be bonded to a temporary support wafer using an adhesive, to handle the wafer for front-side **106** processing. At block **412**, the silicon support wafer is removed from the first surface **106** of the wafer, using one or more techniques, including etching, dissolving, grinding, or the like. The removal process is performed to ensure that the first surface **106** retains the dielectric layer and any conductive layers for future direct or hybrid bonding after the support wafer has been removed.

(57) At block **414**, one or more protective coatings may be applied to the exposed first surface **106**.

In some embodiments, more than one type of protective layer may be applied to the first surface

**106**. For example, a first protective layer may comprise a hydrophobic protective layer and overlaying the hydrophobic layer may be a hydrophilic protective layer (as described above).

(58) At block **416**, the temporary carrier and temporary bonding adhesive are removed from the back surface **110** of the wafer. At block **418**, the finished double-sided wafer is then mounted to a dicing tape on a frame and singulated to form a quantity of double-sided dies **102** (block **420**). Optionally, the diced wafer may be transferred to a grip ring (block **422**) in preparation for bonding (block **424**).

(59) Referring to FIG. 5, in an alternate embodiment, illustrated by the process **500**, the silicon carrier is not bonded to the wafer using a direct bonding technique. Instead, after fabricating (block **502**) and forming/finishing the first bonding surface (block **504**) as described above with reference to FIG. 4, the first surface **106** is bonded to a supporting carrier using a temporary bonding adhesive (block **506**) for fabrication of the second (back side) **110** bonding surface.

(60) At block **508**, the wafer may be thinned as desired, exposing the TSVs. The second bonding layer **110** is formed and finished as described above. At block **510**, one or more protective coatings may be applied to the prepared bonding surface(s) (**106** and/or **110**) prior to dicing. After forming and finishing the second bonding surface **110**, the temporary carrier and temporary adhesive material is removed (block **512**). The finished double-sided wafer is then mounted to a dicing tape or dicing layer on a frame (block **514**), for example, and singulated (block **516**) to form a quantity of double-sided dies **102**. Optionally, the diced wafer may be transferred to a grip ring (block **518**) in preparation for bonding (block **520**).

(61) Cleaning and Preparing the Substrate

(62) The illustrations of FIGS. 6A and 6B show example processes **600** and **650** respectively, describing cleaning and preparing the substrate **104** (process **600**) and the dies **102** (process **650**) for bonding. Again, although the illustrations (and the associated descriptions) describe bonding dies **102** to a substrate **104**, the process can be used to bond wafers to wafers, dies to dies, dies to wafers, and so forth. Moreover, the substrate **104** described in these processes can also refer to the back side **110** of a die **102** already mounted to another die **102** or to a larger substrate, such as a wafer for example.

(63) Referring to FIG. 6A, a process **600** is shown for preparing the bonding surface **108** of the substrate **104**. In some embodiments, the process **600** follows or is summarized within block **206** of the process **200** illustrated at FIG. 2. For instance, at block **602**, the bonding surface **108** of the substrate **104** is cleaned, for example, with a wet process with solvent. Block **604** follows, including a dry clean process with an oxygen plasma (commonly referred to as ashing) or other plasma chemistry, to remove any inorganic and organic contaminants. The plasma may be provided through an atmospheric, downstream, or reactive ion etching (RIE) process, for instance.

(64) The substrate **104** may then be re-cleaned, including scrubbed with deionized or mildly alkaline water or other suitable solution to remove any particles from the ashing process. At block **606**, the bonding surface **108** is activated using a nitrogen plasma process, for example, to prepare the substrate surface **108** for bonding. In various examples, the activation process is known to improve bond strength of the bonded die **102** to the substrate **104**. Similarly, the activation step may be provided through an atmospheric, downstream, or reactive ion etching (RIE) process, for example. The substrate **104** may also be rinsed with deionized water to remove particles from the activation process, as shown at block **608**.

(65) Cleaning and Preparing the Dies

(66) Referring to FIG. 6B, a process **650** is shown for preparing the first bonding surface **106** of the die(s) **102**. In some embodiments, the process **650** follows or is summarized within blocks **316**, **424**, or **520** from FIGS. 3, 4, and 5, respectively, or the like. For instance, at block **652**, preparing the first bonding surface **106** of the dies **102** includes removing the protective layer from the bonding surface **106**. At blocks **654** and **656**, the bonding surface(s) **106** are cleaned. For example,

one or more alternate or optional process stages, including a wet cleaning with a solvent, a dry cleaning with an oxygen plasma (commonly referred to as ashing) or the use of other plasma chemistry, or the like, may be used to remove any inorganic and organic contaminants. The dry cleaning may be performed by a plasma process such as an atmospheric, downstream, or RIE process.

(67) In some cases, the protective layer(s) may be cleaned from the dies **102** while the dies **102** are held on the dicing tape by the frame or grip ring (via a wet process, for example), which also cleans debris resulting from the dicing process. When a UV release tape is used, UV exposure to reduce adhesion strength between the die **102** and the tape to release the die **102** can be performed either before the die cleaning process, or after the die cleaning process, if desired.

(68) In various embodiments, the dies **102** are cleaned with chemicals (e.g., solvents, etchants, water, etc.) that can remove the protective coatings without corroding the metallic layer (e.g., copper) underneath. For example, the cleaning chemicals may include mixtures of hydrogen peroxide in a solution of ammonium hydroxides, or acids. Metal passivating compounds (e.g. triazole moieties) may be used to inhibit metal etching of conductive features such as silver, copper, nickel, and their various alloys. These chemicals and passivating compounds may be cleaned or rinsed from the metal portions of the bonding surface **106**.

(69) During the dicing process, particles of the wafer, for example silicon particles, can be embedded in the dicing tape by the mechanical action of the dicing wheel or wire. These embedded particles may contaminate edges and surfaces of the cleaned dies **102**, including the bonding surfaces **106**, during pick and place direct die transfer from the dicing tape to the bonding layer. Accordingly, it is very important to reduce or eliminate embedded particles from the cleaned bonding surfaces **106**. In one example, the dies **102** may be re-mounted onto a sheet of fresh dicing tape of the same or similar material or another suitable material, discarding the original dicing tape. This can be done to remove the embedded contaminants or to enable the use of chemicals that are not compatible with the original dicing tape.

(70) Alternatively, the contaminants trapped in the dicing tape can be removed with more vigorous cleaning of the dies **102**, including megasonic shaking, mechanical brushing, and/or a high pressure wash. The vigorous cleaning, including megasonic shaking, mechanical brushing, and high pressure wash also cleans edges of the dies **102** to remove particles and organic contaminants at the edges.

(71) The dies **102** may be dry cleaned with a plasma process such as oxygen plasma and re-cleaned with a wet process while on the dicing tape held by the frame or grip ring, to remove any remaining residue of the protective layer(s) or additional contaminants resulting from some process steps, and to improve the bonding surface **106**. However, in some cases, the dicing tape can react with the plasma process and can potentially cause re-contamination of the bonding surface **106** by its reaction products.

(72) In some cases, shielding the exposed tape beyond the die **102** and minimizing the exposed tape between the dies **102** (e.g., using a narrow cutting blade and not stretching the tape) and shortening the plasma process duration can reduce the contamination. In some cases, an oxygen-based reactive ion etching (RIE) plasma process is preferred to shorten the ashing process. In various implementations, a short ashing process in a RIE plasma chamber can cause less surface re-contamination than a long ashing process in a less powerful plasma chamber. The re-cleaning can comprise rinsing the dies **102** with deionized water, which can be combined with mechanical brushing, megasonic shaking, and/or a high pressure wash.

(73) At block **658**, the dies **102** may be activated. A nitrogen-based RIE process or downstream plasma method or other plasma cleaning steps including atomic layer cleaning methods can be used to activate the die surface **106** and/or clean any remaining residue or undesirable materials from the surfaces of interest. At block **660**, as an option, the dies **102** may be rinsed with deionized water for a final cleaning.

(74) At the end of the die **102** forming and preparation process, the dies **102** may be bonded to the prepared bonding surface **108** of the substrate **104** (as illustrated in FIG. **1B**). For a multiple-die **102** stacking arrangement as shown in FIG. **1C**, after placing each die **102A** on the substrate **104** (or previous die **102N**), the second bonding surface **110** (backside) of the placed die **102A** may be planarized, cleaned, and activated (in some embodiments as needed and as discussed above) before direct bonding a bonding surface **106** of a next die **102B** to the backside surface **110** of the previous die **102A**. Any added die **102N** may have a prepared bonding surface **106** and/or **110**, one or both of which may or may not be activated.

(75) In some cases, the nitrogen-based plasma process can be very reactive with the dicing tape and cause bonding surface contamination, which can negate the benefits of plasma activation. Plasma processes (as used in ashing and activation) can also react with the metal device layer on a die **102** or substrate **104** and change the recess of the metal layer from the dielectric surface. Some of the compounds formed as a result of the interaction of the nitrogen plasma with the polymeric dicing tape may adsorb on the bonding surfaces of interest and contaminate the bonding surfaces (**106**, **108**, and in some embodiments **110**).

(76) Consequently, in some embodiments, at least one of the bonding surfaces **106** and/or **110** may not be activated. In some embodiments, a more gentle oxygen or nitrogen plasma of various combinations is preferred to modify the surfaces (**106**, **108**, and in some embodiments **110**) when dicing tapes are present or in close proximity. Similarly, the activated surface (**106**, **108**, and in some embodiments **110**) may be further cleaned. The cleaning step may comprise a rinse with deionized water, which may be combined with mechanical brushing, megasonic cleaning, and/or a high pressure wash.

(77) In various embodiments, some die process steps may be modified or eliminated, in comparison to the process steps described herein.

(78) Cleaning and Activating Dies on the Pick and Place Tool

(79) In an embodiment, a die **102** is selected by pick and place tools (generally comprising a pick tool and a bonding or placement tool), which may take the die **102** through a cleaning process on the way to placing the die **102** on the prepared substrate **104** for bonding. In an implementation, the dies **102** are activated while on the pick and/or place tools. Such cleaning and activating steps performed while the die **102** is on the tool may be in addition, or as an alternative to, certain of the processes described above, though certain details of the steps described above may be performed in a similar fashion.

(80) As an example, the die **102** may be carried by the pick and place tools through a wet cleaning station, an atmospheric plasma, a laser explosion, or the like, on the way to the substrate **104**. In some embodiments, the dies **102** are not rinsed after activation and prior to bonding. In other embodiments, the dies **102** may be rinsed with deionized water, a prepared formulary, or the like, at a process step prior to bonding. In some embodiments, the dies **102** are picked from the dicing tape and flipped by the pick tool to present the back surface **110** of the die **102** to the bonding tool. The face of the pick tool may be in direct contact with the surface **106** to be bonded. The bonding tool head couples to the back side **110** of the die **102**, and brings the prepared surface **106** of the die **102** to the prepared surface **108** of the substrate **104** for bonding.

(81) In still other embodiments, the prepared surface **106** of known good dies **102** are picked from the dicing tape and flipped by the pick tool to present the back side **110** of the die **102** to the bonding tool. The face of the pick tool may be in direct contact with the surface **106** to be bonded. The bonding tool head couples to the back side **110** of the die **102** and brings the prepared face **106** of the die **102** (such as die **102B** at FIG. **1C**) to the exposed prepared surface **110** of known good dies **102** (such as die **102A** at FIG. **1C**) stacked and bonded to the substrate **104** for bonding.

(82) For instance, after cleaning and activating the second surface **110** of the first die **102A** and the first surface **108** of the substrate **104**, the prepared first surface **106** of second known good dies **102B** may be similarly picked from the dicing layer and bonded to the prepared second surface **110**

of the first die **102A**. Multiple known good dies **102 (A-N)** may be stacked over the substrate **104** or over one another, or combinations thereof.

(83) In another embodiment, the prepared dies **102** are picked from the dicing tape and flipped by the pick tool to present the back side **110** of the die **102** to the bonding tool. The bonding tool head couples to the back side **110** of the die **102**, and an electrostatic device may be used to remove particulates from the bonding surface **106** prior to attaching the prepared face **106** of the die **102** to the prepared surface **108** of the substrate **104** for bonding. Moreover, the pick and place tool may have a porous surface for contacting the bonding surface.

(84) In other examples, carbon dioxide particles or compressed gas, for example nitrogen, may be applied at a glancing angle to the prepared surface **106** to remove spurious particulates from the prepared surface **106** prior to bonding. In some applications hot compressed gas, for example, hot nitrogen may be applied to remove spurious particles or excess moisture from the prepared surfaces **106** just before the bonding operation. The pressure of the compressed nitrogen may range between 20 to 300 psi and preferably between 50 to 150 psi. Similarly, the temperature of the compressed gas, for example nitrogen, may range between 25 to 100° C. and preferably between 5° and 90° C. The fluid surface cleaning time may range between 2 to 1000 milliseconds and preferably under 200 milliseconds.

(85) Activating One Bonding Surface

(86) In most of the examples above, the prepared surface **108** of the substrate **104** and the prepared surface **106** of the dies **102** are each plasma cleaned (ashing with oxygen plasma is the most common process) and activated prior to bonding the dies **102** to the substrate **104**. In certain embodiments, however, just one of these bonding surfaces (e.g., the prepared surface **108** of the substrate **104** or the prepared surface **106** of the dies **102**) is plasma cleaned and/or activated prior to bonding the dies **102** to the substrate **104**.

(87) In the embodiments, eliminating surface ashing and activation process on either the dies **102** or on the substrate **104** can decrease process related defects and improve yields, as well as reduce costs for forming the bonding surfaces. For instance, eliminating the process steps can result in eliminating contamination of the bonding surface **106** by the dicing tape/plasma interaction product and reducing or eliminating metal loss from wiring layers. Additionally, damage to the dicing frame or grip ring can be minimized so the frame or grip ring life can be extended. Further, throughput can be improved with fewer process steps taken, and with no reduction in bond energy when activating one surface (**106**, **108**) instead of both surfaces (**106**, **108**). Total cost of ownership is reduced with the elimination of plasma chambers for die **102** cleaning as well.

(88) Cleaning and Preparing the Substrate—Alternate Embodiment

(89) The illustrations of FIGS. 7A and 7B show example processes **700** and **750** respectively, describing alternate processes for cleaning and preparing the substrate **104** (process **700**) and the dies **102** (process **750**) for bonding. Again, although the illustrations (and the associated descriptions) describe bonding dies **102** to a substrate **104**, the process can be used to bond wafers to wafers, dies to dies, dies to wafers, and so forth. Moreover, the substrate **104** described in these processes can also refer to the back side of a die **102** already mounted to another die **102** or to a larger substrate, such as a wafer.

(90) Referring to FIG. 7A, a process **700** is shown for preparing the bonding surface **108** of the substrate **104**. At block **702**, the substrate **104** is fabricated, and a bonding surface **108** is formed on the substrate **104** at block **704**, in like manner as described with regard to process **200** at FIG. 2. For example, the bonding surface **108** is prepared for direct bonding without adhesive. In some embodiments as disclosed earlier, fabrication of the substrate **104** may also include forming the bonding surface **108**.

(91) At block **706**, the bonding surface **108** of the substrate **104** is cleaned, for example, with a wet process with solvent, or the like. Block **708** follows, including a dry clean process with an oxygen plasma (commonly referred to as ashing) or other plasma chemistry, to remove any inorganic and

organic contaminants. The plasma may be provided through an atmospheric, downstream, or reactive ion etching (RIE) process, for instance. The substrate **104** may be re-cleaned, including scrubbed with deionized or mildly alkaline water or other suitable solution to remove any particles from the ashing process. In some embodiments, the bonding surface **108** of the substrate **104** may be rendered hydrophobic by cleaning methods, before storing the substrate **104** for subsequent use or before the activation step. At block **710**, the bonding surface **108** is activated using a nitrogen plasma process, for example, to prepare the substrate surface **108** for bonding. In various examples, the activation process is known to improve bond strength. The substrate **104** may also be rinsed with deionized water to remove potential contaminating particles from the activation process, if desired.

(92) Cleaning and Preparing the Dies—Alternate Embodiment

(93) Referring to FIG. 7B, a process **750** is shown for preparing the bonding surface(s) **106** and/or **110** of the die(s) **102**. At block **752**, the dies **102** are fabricated on a wafer, and at block **754** a bonding surface **106** is formed and prepared on a surface of the dies **102**, as described previously with regard to process **300** at FIG. 3. For example, the bonding surface **106** is prepared for direct bonding without adhesive. At block **756** one or more protective layers may be added to the bonding surface **106**, as also described above, and at block **758**, the wafer is singulated into a plurality of dies **102**.

(94) At block **760**, the process includes removing the protective layer(s) from the bonding surface **106**. At block **762**, the bonding surface(s) **106** are cleaned using a chemical formulary, as described further below. In some cases, the protective layer(s) may be cleaned from the dies **102** while the dies **102** are held on the dicing tape by the frame or grip ring, which also cleans debris resulting from the dicing process. When a UV release tape is used, UV exposure to reduce adhesion strength between the die **102** and the tape to release the die **102** can be performed either before the die cleaning process, or after the die cleaning process, if desired.

(95) Advantages to the process **750**, including cleaning (or other processing) the dies **102** while on the frame or grip ring, or on a polymeric sheet include: elimination of polymeric residue from oxygen and nitrogen plasma steps; reduced processing steps and cycle times for manufacturing; the dies **102** may not need to be activated when the substrate **104** has been activated; the bond energy of the process (including without the die activated) is comparable to a like process where the dies **102** are activated along with the substrate **104** activation; and higher throughput of manufactured devices.

(96) Example Chemical Formulary

(97) Referring to FIG. 7B, after the protective layer is cleaned from the singulated dies **102** (and optionally after the dies **102** are cleaned with a plasma process) the dies **102** may be exposed to a chemical formulary for a predetermined time to clean and prepare the die surface(s) **106** and/or **110** for bonding. In the embodiment, the prepared formulary comprises glycerated diluted hydrofluoric acid or buffered hydrofluoric acid or ammonium fluoride, organic acid, and deionized water, with or without a stabilizing additive. In some embodiments, the formulary may comprise an inorganic or organic acid containing a fluoride ion. It is preferable that the content of the fluoride ion be less than 2% and preferably less than 0.5% and, in some cases, preferably less than 0.1%. Examples of the sources of fluoride ions may include hydrofluoric acid, buffered oxide etch, ammonium fluoride, or tetrabutylammonium fluoride.

(98) The formulary may also comprise aliphatic or non-aliphatic organic acids, and more than one organic acid may be used in the formulary. The organic acid content of the formulary may typically be less than 2% and preferably less than 1%, and preferably less than 0.1%. Examples of organic acid may include formic acid, acetic acid, methyl sulfonic acid and their likes. In some embodiments, mineral acids (for example, a very small amount of sulfuric acid) may be used. However, the amount used should not roughen the surface of the metallic layer at the bonding surface **106**.

(99) In various embodiments, glycerol is incorporated into the formulary, where the content of glycerol may vary between 0.5 to 25% of the formulary, and preferably under 10%. In other applications, a very small amount of amide, amines, butylated hydroxyanisole (BHA), butylated hydroxytoluene, or organic carbonates may be added to the formulary. In other embodiments, the formulary may be mildly alkaline with a pH preferably less than 9.5 and preferably less than 8.5. It is preferable that the total content of these additional additives be less than 5% and preferably less than 1%.

(100) It is also desirable that a complexing agent that suppresses the removal or etching or roughening of the surface of the metallic layer at the bonding surface **106** be incorporated into the formulary. In the case of copper, for example, a suitable copper complexing agent with one or more triazole moieties may be used. The concentration of the complexing agent may be less than 2%, and preferably less than 1%, 0.2%, and less than 100 ppm and less than 5 ppm in some instances. After exposure to the formulary, the dies **102** are placed by the pick tool and bonded to the substrate **104**, without activation (and optionally without ashing).

(101) After cleaning with the formulary, the dies **102** may be rinsed (for example with deionized water), and may be activated. A nitrogen-based RIE process or downstream plasma method or other plasma cleaning steps including atomic layer cleaning methods can be used to activate the die surface **106** and/or clean any remaining residue or undesirable materials from the surfaces of interest. As an option, the dies **102** may be rinsed with deionized water after activation.

(102) At the end of the die **102** forming and preparation process, the dies **102** may be bonded to the prepared bonding surface **108** of the substrate **104** (as illustrated in FIG. 1B). For a multiple-die stacking arrangement as shown in FIG. 1C, after placing each die **102A** on the substrate **104** (or another die **102N**), the second bonding surface **110** (backside) of the placed die **102A** may be planarized, cleaned, and activated (as needed and as discussed above) before direct bonding a next die **102B** to the backside surface **110** of the previous die **102A**. Any added die **102** (A-N) may have a prepared bonding surface **106** and/or **110**, which may or may not be activated.

(103) Forming a Hydrophobic Bonding Surface

(104) In another embodiment, the surface of the dies **102** or substrate **104** or both may be rendered hydrophobic by one or more of the cleaning steps described above. One advantage of rendering the surface (**106**, **108**, and/or **110**) itself hydrophobic is that hydrophobic surfaces may be less prone to particulate contamination, and are easily cleaned with compressed fluids such as nitrogen, carbon dioxide or carbon dioxide particles. The bonding surface **106** or **110** of the die **102** (while in the dicing frame, for instance) or the bonding surface **108** of the substrate **104** may be rendered hydrophobic by exposing their non-activated surfaces (**106**, **108**, and/or **110**) to the very dilute fluoride containing formularies described above, then rinsing and drying the exposed surface (**106**, **108**, and/or **110**). According to this technique, a cleaned, unactivated die **102** with a hydrophobic surface **106** and/or **110** may be attached to the surface **108** of a prepared, activated substrate **104** (or to another die **102**) for bonding. Similarly, a cleaned, unactivated substrate **104** with a hydrophobic surface **108** may be bonded to the surface **106** of an activated die **102** or dies **102**. Generally, nitrogen activation of a surface (**106**, **108** and/or **110**) tends to render the surface (**106**, **108** and/or **110**) hydrophilic. Annealing the mated surfaces (**106**, **108** and/or **110**) improves the bond energy between the mated materials. Generally, the higher the bonding temperature, the higher the energy need to separate the mated materials.

(105) Additional Process Step Reductions

(106) In a further embodiment, after the protective layer is cleaned from the singulated dies **102**, the dies **102** are placed by the pick and place tools and bonded to the substrate **104**, without exposure to the formulary, ashing, or activation. In various embodiments, with the substrate **104** activated and the die **102** not activated, the bond energy (or the strength of the bond) of the resulting direct bond is sufficient for the DBI formation process. For instance, in some embodiments, the bond energy is approximately 1000 mJ/m<sup>2</sup>, which meets minimum bond

energy requirements for a suitable direct bond according to some specifications. Further, elimination of certain oxygen and nitrogen RIE steps eliminates the dicing tape/plasma reaction by-products which can deposit on the die **102** surface, as well as the cost and time of additional process steps.

#### (107) Annealing

(108) As a final step in the bonding process for all of the embodiments discussed, the die **102** and substrate **104** may be heated above ambient temperature (annealed) to form the metal-to-metal joint. In these embodiments, the high temperature of the annealing process causes the metal (e.g., Cu) in the bonding layer (**106**, **108** and/or **110**) to expand more than the dielectric material (e.g. oxide) surrounding the metal. The differential in CTE allows the metallic layer(s), which may be recessed below the bonding surface (**106**, **108** and/or **110**) at room temperature, to expand to bridge the gap between the two mating surfaces of the dielectric materials and form electrically conductive joints during the anneal.

(109) In one embodiment, a multi-die stack is not annealed after each die **102** is placed, but instead the whole stack is annealed once all of the dies **102** of the stack are placed. Alternatively, a low temperature anneal may be used after each die **102** is placed. Further, the back surface **110** of the bonded die **102** is cleaned and prepared to accept an additional cleaned die **102** or die stack. The substrate **104** with a multi-die stack may be thermally annealed at a higher temperature for the opposing metal features at the various bonding interfaces to mate intimately. In some embodiments, after disclosed processes, the substrate **104** may be singulated by known methods to form a new structure comprising of dies **102**, directly bonded to a smaller singulated substrate **104** (not shown). In one embodiment, a bonding surface area of a die **102** is smaller than the bonding surface of the singulated substrate **104**.

#### (110) Pick and Place Tool Preparation

(111) In various implementations, the pick and place tools are designed or treated to minimize the opportunities to contaminate the dies **102**. The pick and place tools (or portions of them) may be formed from a material (or coated with a material) selected for desired hydrophobic properties. For instance, the tools may be coated with a material such as polytetrafluoroethylene (PTFE), or another hydrophobic material.

(112) Also, the tools may be prepared (e.g., with a structural design, with a predetermined treatment, etc.) to be resistant to contaminants or to avoid passing contaminants to the dies **102**. As such, the face pick process (e.g., picking the dies **102** at the prepared bonding surface **106** and/or **110**) does not degrade bonding quality while picking from the die surface **106** and/or **110** to be bonded. Alternatively, the surface **106** and/or **110** of the die **102** can be chemically treated to have hydrophobic properties to minimize particles sticking to the surface **106** and/or **110** from the pick tool. Also, at desirable intervals, the surface of the pick and place tool can be cleaned to remove potential sources of die surface contaminants. One or more surfaces of the tool may be porous.

(113) The techniques, components, and devices described herein are not limited to the illustrations of FIGS. **1A-7B**, and may be applied to other designs, types, arrangements, and constructions including with other electrical components without departing from the scope of the disclosure. In some cases, additional or alternative components, techniques, sequences, or processes may be used to implement the techniques described herein. Further, the components and/or techniques may be arranged and/or combined in various combinations, while resulting in similar or approximately identical results.

#### CONCLUSION

(114) Although the implementations of the disclosure have been described in language specific to structural features and/or methodological acts, it is to be understood that the implementations are not necessarily limited to the specific features or acts described. Rather, the specific features and acts are disclosed as representative forms of implementing example devices and techniques.



## Claims

1. A bonded structure comprising: a substrate directly hybrid bonded to a die along a bonding interface without an intervening adhesive, the die having a smaller footprint than a footprint of the substrate, wherein the substrate comprises one or more metal features, wherein the die comprises one or more metal features, and wherein the one or more metal features of the substrate are directly bonded to the one or more metal features of the die at the bond interface, wherein one of the die and the substrate comprises a characteristic profile of having been activated with an activating species prior to bonding, and the other of the die and the substrate lacks the characteristic profile of having been activated with the activating species prior to bonding.
2. The bonded structure of claim 1, wherein the activating species comprises nitrogen.
3. The bonded structure of claim 1, wherein the substrate is directly bonded to the die with a hybrid bond having a bond energy of at least 1000 mJ/m.<sup>2</sup>.
4. The bonded structure of claim 1, wherein the substrate comprises a first insulating region that at least partially surrounds the one or more metal features of the substrate, wherein the die comprises a second insulating region that at least partially surrounds the one or more metal features of the die, and wherein the first insulating region is directly bonded to the second insulating region without an adhesive.
5. The bonded structure of claim 4, wherein the first insulating region is directly bonded to the second insulating region with a chemical bond.
6. A method of forming a microelectronic assembly, the method comprising: providing a substrate having a first bonding surface, wherein the first bonding surface comprises a first metal feature; providing a die having a second bonding surface, wherein the second bonding surface comprises a second metal feature; plasma activating only one of the first bonding surface and the second bonding surface; and hybrid bonding the second bonding surface to the first bonding surface without an adhesive and without activating the other of the first bonding surface and the second bonding surface, wherein, after hybrid bonding the second bonding surface to the first bonding surface, the first and second metal features are directly bonded together.
7. The method of claim 6, wherein plasma activating only one of the first bonding surface and the second bonding surface comprises activating only one of the first bonding surface and the second bonding surface with a nitrogen plasma.
8. The method of claim 6, further comprising: before plasma activating only one of the first bonding surface and the second bonding surface, cleaning the only one of the first bonding surface and the second bonding surface.
9. The method of claim 8, wherein cleaning the only one of the first bonding surface and the second bonding surface comprises plasma cleaning the only one of the first bonding surface and the second bonding surface.
10. The method of claim 9, wherein plasma cleaning the only one of the first bonding surface and the second bonding surface comprises cleaning the only one of the first bonding surface and the second bonding surface with oxygen plasma.
11. The method of claim 9, wherein plasma cleaning the only one of the first bonding surface and the second bonding surface comprises exposing the second bonding surface to oxygen plasma without exposing the first bonding surface to oxygen plasma.
12. The method of claim 6, wherein: the first bonding surface comprises a first conductive feature and a first insulating region at least partially surrounding the first conductive feature, the second bonding surface comprises a second conductive feature and a second insulating region at least partially surrounding the second conductive feature, and hybrid bonding the first bonding surface to the second bonding surface comprises directly hybrid bonding the first bonding surface to the second bonding surface such that the first conductive feature is directly bonded to the second

conductive feature and the first nonconductive region is directly bonded to the second nonconductive region.

13. The method of claim 6, wherein hybrid bonding the second bonding surface to the first bonding surface without an adhesive comprises forming a hybrid bond having a bond energy of at least 1000 mJ/m.<sup>2</sup>.

14. A method of forming a microelectronic assembly, the method comprising: preparing a bonding surface of a first substrate; preparing a bonding surface of a second substrate; applying a protective coating to the bonding surface of the second substrate; singulating the second substrate into a plurality of dies, wherein the plurality of dies comprises a first die, wherein the first die has a bonding surface that comprises a portion of the bonding surface of the second substrate and has a portion of the protective coating; plasma activating the bonding surface of the first substrate; cleaning off the portion of the protective coating from the first die to expose the bonding surface of the first die; and directly bonding the bonding surface of the first die to the bonding surface of the first substrate without an adhesive, wherein the bonding surface of the first die is not activated before it is directly bonded to the bonding surface of the first substrate.

15. The method of claim 14, wherein plasma activating the bonding surface of the first substrate comprises plasma activating the bonding surface of the first substrate with a first plasma.

16. The method of claim 15, wherein the first plasma comprises a nitrogen plasma.

17. The method of claim 15, wherein cleaning off the portion of the protective coating from the first die to expose the bonding surface of the first die comprises cleaning off the portion of the protective coating from the first die with a non-activating second plasma that is different from the first plasma.

18. The method of claim 17, wherein the second plasma comprises an oxygen plasma.

19. The method of claim 17, wherein the bonding surface of the first die is not exposed to the second plasma.

20. The method of claim 14, wherein cleaning off the portion of the protective coating from the first die to expose the bonding surface die comprises cleaning off the portion of the protective coating from the first die with a reactive ion etching process.

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